

ECE 473/573
Cloud Computing and Cloud Native Systems
Lecture 05 Virtualization

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Outline

Virtual Machines

CPU and Memory Virtualization

Reading Assignment

- ▶ This lecture: Virtualization
- ▶ Next lecture: Containerization

Outline

Virtual Machines

CPU and Memory Virtualization

Physical Server Operation and Maintenance

- ▶ Ensure health and security of physical servers.
 - ▶ Configure storage and networking.
 - ▶ OS/library installation and updates.
 - ▶ What about doing so for thousands of servers?
- ▶ Usually access to physical servers is very limited.
 - ▶ Avoid accidental damage to other servers, power delivery, cooling solutions, and network connections.
 - ▶ Security concerns for physical accesses.
- ▶ Most if not all servers support remote accesses via Intelligent Platform Management Interface (IPMI).
 - ▶ KVM (Keyboard, Video and Mouse) over network.
 - ▶ Run on an embedded computer inside the server independent of host OS, and can be automated via scripting.
 - ▶ Make security concerns more complicated.

Application Operation and Maintenance

- ▶ Applications depend on OS/library to work properly.
 - ▶ May need specific versions of such dependencies.
 - ▶ Dependencies have their dependencies as well.
- ▶ Running a single application per server is not efficient. But if we run multiple applications together in the same os,
 - ▶ Dependency conflicts may arise when two applications require two different versions of the same library.
 - ▶ Security issues due to bugs or misconfigurations in OS, library or one application may propagate to other applications.
- ▶ Dependency issues can be solved by languages themselves.
 - ▶ With good backward compatibility like Java
 - ▶ With version management like Python
- ▶ Security concerns are difficult to address.

Virtual Machines

- ▶ Cloud computing utilizes virtual machines instead of accessing physical servers directly.
 - ▶ Multiple virtual machines can execute simultaneously on a single physical server.
 - ▶ Each with a guest OS, which can support many applications
- ▶ Virtual machines need both hardware and software supports.
 - ▶ Hardware: processor, memory system, and I/O devices
 - ▶ Software: hypervisor

Hypervisor

- ▶ A software system to manage multiple VMs.
 - ▶ Emulate hardware resources so guest OSES and applications can execute without modification.
 - ▶ Allocate and ensure fair using of hardware resources.
 - ▶ Isolate failures and prevent interference and security breaches between VMs.
- ▶ Hypervisors usually work in two ways.
 - ▶ Type 1 (Bare-Metal): installed directly on the physical hardware to provides direct access to hardware resources.
 - ▶ Type 2 (Hosted): installed on top of a host OS and relies on it for resource management.
 - ▶ Type 1 hypervisors usually have better performance for production use while type 2 hypervisors are more flexible for development and testing.

Isolation: OS vs Hypervisor

- ▶ Modern OS provides mechanisms for isolation.
 - ▶ Leverage CPU rings (privilege levels) to protect itself.
 - ▶ Applications run in their own virtual memory space so memory access faults are isolated.
 - ▶ Users and ACLs (access control lists) are used to controls accesses to resources like files.
- ▶ However, OS need to provide rich functionalities.
 - ▶ More chances to have bugs and misconfigurations.
 - ▶ Breaking security guarantees.
- ▶ Hypervisors focus on emulation, allocation, and isolation.
 - ▶ Much smaller than a full featured OS.
 - ▶ Less chances to have bugs and misconfigurations.
 - ▶ How could applications running in different VMs interfere with each other if they are not aware of each other?

Outline

Virtual Machines

CPU and Memory Virtualization

CPU Virtualization

- ▶ Each VM uses a predefined number of cores.
 - ▶ In addition, VM may have CPU performance requirement, usually measured as clock frequency.
- ▶ Hypervisor allocates enough cores to a VM and let it run for a short period of time.
 - ▶ At a frequency meeting the clock frequency requirement.
 - ▶ E.g. for a requirement of 1GHz on a 2GHz physical core, run for a total 0.5 second for every second.
 - ▶ Use a scheduling algorithm like round-robin when there are multiple VMs.
- ▶ For example, consider a physical server with 16 2GHz cores.
 - ▶ 1 VM requiring 16 1GHz cores, 1 VM requiring 8 1GHz cores, 2 VMs requiring 4 1GHz cores.
 - ▶ To simplify allocation and scheduling, number of cores used by VMs are usually powers of 2.

Handling Interrupts

- ▶ Interrupts are handled by individual cores.
- ▶ The hypervisor sets up interrupt handles (ISR) so that it can catch these events and take control.
 - ▶ Decide what OS it belongs to.
 - ▶ Make timely VM scheduling decisions.
- ▶ Software interrupts are usually due to system calls.
 - ▶ The hypervisor forwards them to the OS.
- ▶ External interrupts are usually from I/O devices.
 - ▶ Depend on different device virtualization mechanisms, the hypervisor sends a corresponding one to the OS.
- ▶ Faults like null pointer exception and page faults.
 - ▶ Handled in a similar way as software interrupts.

Overprovisioning

- ▶ What if the total cores and clock frequencies required by all the VMs exceeds what the physical server can provide?
- ▶ Assumption: CPU demands from VMs fluctuations over time.
 - ▶ Not all VMs will be using cores at their peak at the same time.
- ▶ Overprovisioning improves resource utilization.
 - ▶ By packing more VMs into a physical server, it is more likely to have some VMs using cores instead of all cores being idle.
- ▶ Higher resource utilization translates to lower cost.
 - ▶ Enable incentives for clients to relax their requirements for lower cost, e.g. no frequency requirement at all, which
 - ▶ makes overprovisioning even more attractive as there is no need to meet all the core/frequency demands 100% of the time.
- ▶ Careful planning and monitoring are still necessary to meet requirements that are not relaxed.

Memory Virtualization

- ▶ OS, together with CPU, provides mapping between physical memory to virtual memory that an application can use.
 - ▶ When that is not sufficient physical memory, virtual memory is backed by disk storage at a much lower performance.
 - ▶ OS and applications may optimize for such scenarios as they may have knowledge of so.
- ▶ For guest OS running inside a hypervisor, memory virtualization involves host memory (machine memory), guest memory (physical memory), and virtual memory.
 - ▶ If there is not enough host memory, backing guest memory via disk storage is not ideal: guest OS and applications have no knowledge of such and cannot optimize for it.
- ▶ When leveraging memory overprovisioning to improve host memory utilization, there is always not enough host memory.
 - ▶ Can the hypervisor notify guest OS of desired memory usages without changing how guest OS works?

Summary

- ▶ Hypervisor manages VMs on a single server.
- ▶ Hypervisors are less complicated than OS, and are less likely to have bugs and misconfigurations.
- ▶ Overprovisioning optimizes resource usages and improves resource utilization.